

Lecture 21: Matching via Augmenting Paths

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Consider the following problems:

1. Independent Set (IND SET)

Given a graph $G = (V, E)$, find a maximum independent set:

$$S \subseteq V \text{ such that } \forall u, v \in S, (u, v) \notin E.$$

That is, no two vertices in S are adjacent.

2. Clique

Given a graph $G = (V, E)$, find a maximum clique:

$$S \subseteq V \text{ such that } \forall u, v \in S, (u, v) \in E.$$

That is, every pair of vertices in S is connected by an edge.

3. 3-Coloring (3-COL)

Given a graph $G = (V, E)$, determine whether it is 3-colorable:

$$C : V \rightarrow \{1, 2, 3\} \text{ such that } (u, v) \in E \Rightarrow C(u) \neq C(v).$$

This means adjacent vertices must receive different colors.

Claim: IND SET, Clique, and 3-COL are NP-complete.

Hints for Solving the Problems

Step 1: Convert to Decision Problem

- **IND SET:** Given (G, k) , does G have an independent set of size $\geq k$?
- **Clique:** Given (G, k) , does there exist

$$S \subseteq V \text{ such that } |S| = k \text{ and } \forall u, v \in S, (u, v) \in E?$$

- **3-COL:** Given G , does there exist

$$C : V \rightarrow \{1, 2, 3\} \text{ such that } (u, v) \in E \Rightarrow C(u) \neq C(v)?$$

Step 2: Prove the Problems are in NP

- **IND SET:** Certificate is $S \subseteq V$. Verify:
 - $|S| \geq k$
 - No edge exists between any pair in S
- **Clique:** Certificate is $S \subseteq V$. Verify:
 - $|S| = k$
 - Every pair of vertices in S is connected
- **3-COL:** Certificate is $C : V \rightarrow \{1, 2, 3\}$. Verify:
 - For each edge (u, v) , check $C(u) \neq C(v)$

All verifications run in polynomial time. Hence all are in NP.

Step 3: Prove NP-Hardness

We already know:

$$3\text{-SAT} \leq_p \text{Vertex Cover}$$

So the natural idea is to reduce from Vertex Cover.

Question: Can two adjacent vertices both be outside the vertex cover?

Answer: No. Since every edge must be covered, at least one endpoint of every edge must belong to the vertex cover.

1 Vertex Cover $\leq_p$ Independent Set (IND SET)

- **Claim:** Complement of a vertex cover is an independent set.

Proof:

Let $C \subseteq V$ be a vertex cover. Then for every edge $(u, v) \in E$, at least one of u or v belongs to C .

Hence, no edge can have both endpoints in $V \setminus C$. Therefore, the set $V \setminus C$ contains no adjacent vertices and is an independent set.

Conversely, let $S \subseteq V$ be an independent set. Then no edge exists between any two vertices in S .

Thus, every edge must have at least one endpoint in $V \setminus S$, which implies $V \setminus S$ is a vertex cover. □

- **Reduction:**

$$(G, k) \in \text{VC} \iff (G, n - k) \in \text{IND SET}$$

This transformation can be done in polynomial time, hence

$$\text{VC} \leq_p \text{IND SET}.$$

2 IND SET $\leq_p$ Clique

- **Definition:** The complement graph $\overline{G} = (V, \overline{E})$ is defined by

$$(u, v) \in \overline{E} \iff (u, v) \notin E$$

- **Claim:** A set $S \subseteq V$ is an independent set in G if and only if it is a clique in $\overline{G}$.

Proof:

Assume S is an independent set in G . Then no edge exists between any two vertices in S .

Hence, all such non-edges in G become edges in $\overline{G}$, so every pair of vertices in S is connected in $\overline{G}$. Therefore, S forms a clique.

Conversely, assume S is a clique in $\overline{G}$. Then every pair of vertices in S is connected in $\overline{G}$.

This implies that no such edges exist in G , hence S is an independent set in G . $\square$

- **Reduction:**

$$(G, k) \in \text{IND SET} \iff (\overline{G}, k) \in \text{Clique}$$

Since the complement graph can be constructed in polynomial time,

$$\text{IND SET} \leq_p \text{Clique}.$$

Reduction Chain

$$3\text{-SAT} \leq_p \text{Vertex Cover} \leq_p \text{IND SET} \leq_p \text{Clique}$$

Thus, both the problems are NP-hard.

3 3-Coloring is NP-Complete.

Problem: Given a graph $G = (V, E)$, determine whether it is **3-colorable**, i.e.,

$$C : V \rightarrow \{1, 2, 3\} \quad \text{such that } (u, v) \in E \Rightarrow C(u) \neq C(v).$$

Proof:

Step 1: 3-COL $\in$ NP

Given a coloring C , we verify for each edge $(u, v) \in E$ that:

$$C(u) \neq C(v)$$

This can be done in $O(|E|)$ time. Hence, 3-COL $\in$ NP.

Step 2: NP-Hardness

We reduce from 3-SAT.

Let

$$\phi = C_1 \wedge C_2 \wedge \cdots \wedge C_m$$

where each clause is:

$$C_i = (\ell_{i1} \vee \ell_{i2} \vee \ell_{i3})$$

Construction:

- Create three special vertices TRUE, FALSE, BASE forming a triangle.
- For each variable x , create vertices x and $\neg x$, connect them and also connect both to BASE.
- For each clause $(\ell_1 \vee \ell_2 \vee \ell_3)$, construct a gadget ensuring at least one literal must be TRUE.

Step 3: Correctness

($\Rightarrow$) If ϕ is satisfiable, assign colors TRUE and FALSE accordingly. This extends to a valid 3-coloring.

($\Leftarrow$) If the graph is 3-colorable, extract assignment from variable nodes. Clause gadgets ensure at least one literal is TRUE, hence ϕ is satisfiable.

Conclusion

3-COL $\in NP$ and is NP-hard

$\Rightarrow$ 3-COL is NP-complete

□

4 Maximum Matching in Bipartite Graph

Problem: Given a bipartite graph $G = (L \cup R, E)$, find a maximum matching.

Proof:

Suppose, we have a maximal matching M_1 .

Define:

$$V_1 = \{u, v \mid (u, v) \in M_1\}$$

Let M_{OPT} be a maximum matching, and define:

$$V_{\text{OPT}} = \{u, v \mid (u, v) \in M_{\text{OPT}}\}$$

Again, observe that, for every edge $(u, v) \in M_{\text{OPT}}$, at least one of u or v must be in V_1 .

If both u and v are not in V_1 , then both are unmatched in M_1 . Hence, we can add (u, v) to M_1 , contradicting maximality of M_1 .

Thus,

$$|M_1| \geq \frac{|M_{\text{OPT}}|}{2}$$

Hence, M_1 is a 2-approximation to the maximum matching. □

Homework:-

Show that: $|V_1| \leq 2|V_{\text{OPT}}|$.

Proof:

Since each edge contributes two vertices,

$$|V_1| = 2|M_1|, \quad |V_{\text{OPT}}| = 2|M_{\text{OPT}}|$$

Each edge in M_1 must share at least one vertex with V_{OPT} , since otherwise we could

add that edge to M_{OPT} , contradicting maximality of M_{OPT} .

Thus,

$$|M_1| \leq |V_{\text{OPT}}|$$

Multiplying both sides by 2,

$$|V_1| = 2|M_1| \leq 2|V_{\text{OPT}}|$$

□

Note : If a matching is not maximum, then there exists an augmenting path between two unmatched vertices. So,

Matching is maximum $\iff$ No augmenting path exists

5 Finding Augmenting Path using BFS

Idea

We run BFS on an alternating graph following two simple rules:

- From left side L to right side R , we only travel along **unmatched edges**
- From right side R to left side L , we only travel along **matched edges**

We stop when we reach a free (unmatched) vertex on the right side. The path from start to end is our augmenting path.

Algorithm

Let $G = (L \cup R, E)$ be a bipartite graph and M be the current matching.

Step 1: Setup

- Put all free vertices (unmatched) from L into a queue
- Mark them as visited and set their distance to 0

Step 2: BFS Loop

While queue is not empty:

- Remove the front vertex u from queue
- **Case 1:** u is in L
 - Look at all neighbors v of u through edges **not** in M
 - If v is not visited yet:
 - * Mark v as visited
 - * Record u as the parent of v
 - * Add v to the queue
- **Case 2:** u is in R
 - If u is matched, let v be its partner in M
 - If v is not visited yet:
 - * Mark v as visited
 - * Record u as the parent of v
 - * Add v to the queue

Step 3: Check Result

- If we reached any free vertex in R , then an augmenting path exists
- Otherwise, no augmenting path exists.

□